Soo Yeon Ahn

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Personal Summary

• Professional Pedigree: Research Engineer at Hanwha Aerospace; formerly developed core C modules at Dassault Systèmes supporting 300K+ users
• Technical Expertise: Proficient in low-level, high-performance backend development utilizing object-oriented programming (C/C++)
• Quantitative Edge: Graduated with departmental distinction in Math/CS from UIUC; strong foundation in advanced algorithm design and mathematical proofs to verify system correctness

Work Experience

Hanwha Aerospace — Changwon, Republic of Korea (On-Site)
Research Engineer — Feb 2026 - Current
• Contributed to software architecture planning, focusing on real-time data visualization and intuitive user workflows under high-stress, mission-critical operational conditions
• Drafted system performance expectations and technical specifications, translating complex requirements into comprehensive documentation for cross-functional alignment

Dassault Systèmes — Daegu, Republic of Korea (On-Site)
Software Engineer Intern — Jul 2023 – Jan 2024
• Engineered and maintained C++/COM modules for DELMIA, part of a platform with 300K+ users
• Had full lifecycle ownership of a feature; managing development documentation, development, and regression testing before sending it to the QA team
• Designed and integrated UI components using proprietary frameworks to improve UX
• Collaborated with international teams on cross-border software solutions and code reviews

Education

University of Illinois Urbana-Champaign — Aug 2021 – May 2025
B.S. in Mathematics and Computer Science — Overall GPA: 3.69/4.0
• Achievements: Distinction Math/CS (Department Honors); Fall 2021, Spring 2022 (Dean's List)

Technical Skills

• Computer Languages: C/C++, Python, JavaScript, Java, MIPS
• Databases: MySQL, MongoDB, SQLite
• Version Control: Git (CLI), GitHub, Docker

Teaching & Leadership

Computer Science Department (UIUC) — Jan 2022 – May 2023
Course Assistant (Introduction to Computer Science II) — Urbana, IL
• Led weekly lab sessions (~30 students) on C++ fundamentals and data structure algorithms
• Assisted hands-on debugging and personalized mentoring during 4–6 hours of office hours weekly
• Supported students in developing coding projects and improving problem-solving techniques

Projects

C++ Limit Order Book (Solo) GitHub — December 2025
• Engineered a low-latency matching engine in Modern C++ that handles 930,000 events/second
• Optimized order lifecycle management using red-black trees (std::map) and FIFO queue (std::list)
• System reached 49.84% quantity fill rate under various probabilistic loads
• Standardized build process using CMake to ensure compatibility and streamlined dependency management

Languages

• English (Fluent)
• Mandarin (HSK Level 4 - Intermediate)
• Korean (Native)